Stage 1: Cleaning and Preprocessing Planet Data

In [1]:
```
# This code was created by me using lecture videos from the Data Science course
#and Database and Advanced Data Techniques and previous projects in Data Science li


#load libraries
import pandas as pd
import numpy as np
```

Planet Column Data

In [2]:
```
#load full column data file
column_data=pd.read_csv(r"C:\Users\maybe\OneDrive\Desktop\Databases and Advanced Te

#check first rows to verify it loaded correctly
print(column_data.head(1))
```

```
   column_id abbrev_name  full_name
0          1   planet_id  Planet ID
```

In [3]:
```
#### Find critical columns in column data
critical_column_data=['planet_id','pl_name', 'hostname', 'pl_letter', 'sy_snum', 's
                        'sy_mnum','discoverymethod', 'disc_year', 'disc_facility',
                        'pl_orbper','pl_orbsmax', 'pl_angsep' , 'pl_rade', 'pl_bma
                       'pl_dens', 'pl_orbeccen', 'pl_insol', 'pl_eqt', 'pl_orbincl
                       'pl_trandep', 'pl_trandur', 'pl_rvamp', 'st_spectype', 'st_
                       'st_met', 'st_lum', 'st_logg', 'st_age', 'st_dens', 'st_vsi
                       'st_radv', 'ra', 'dec', 'sy_dist']

new_column_data = column_data[column_data['abbrev_name'].isin(critical_column_data)

#new_column_data = column_data[critical_column_data]
print(new_column_data)
```

```
     column_id      abbrev_name                                  full_name
0            1        planet_id                                  Planet ID
1            2          pl_name                                Planet Name
2            3         hostname                                  Host Name
3            4        pl_letter                              Planet Letter
9           10          sy_snum                            Number of Stars
10          11          sy_pnum                         Number of Planets
11          12          sy_mnum                            Number of Moons
13          14  discoverymethod                          Discovery Method
14          15        disc_year                            Discovery Year
18          19    disc_facility                        Discovery Facility
32          33        pl_orbper                      Orbital Period [days]
37          38       pl_orbsmax                 Orbit Semi-Major Axis [au]
42          43        pl_angsep                    Angular Separation [mas]
47          48          pl_rade                Planet Radius [Earth Radius]
57          58        pl_bmasse   Planet Mass or Mass*sin(i) [Earth Mass]
68          69          pl_dens                    Planet Density [g/cm**3]
73          74      pl_orbeccen                                Eccentricity
78          79         pl_insol                Insolation Flux [Earth Flux]
83          84           pl_eqt                Equilibrium Temperature [K]
88          89       pl_orbincl                          Inclination [deg]
93          94       pl_tranmid                     Transit Midpoint [days]
105        106       pl_trandep                          Transit Depth [%]
110        111       pl_trandur                    Transit Duration [hours]
141        142         pl_rvamp            Radial Velocity Amplitude [m/s]
156        157      st_spectype                               Spectral Type
163        164           st_rad              Stellar Radius [Solar Radius]
168        169          st_mass                  Stellar Mass [Solar mass]
173        174           st_met                  Stellar Metallicity [dex]
179        180           st_lum            Stellar Luminosity [log(Solar)]
184        185          st_logg  Stellar Surface Gravity [log10(cm/s**2)]
189        190           st_age                          Stellar Age [Gyr]
194        195          st_dens                  Stellar Density [g/cm**3]
199        200          st_vsin         Stellar Rotational Velocity [km/s]
204        205          st_rotp           Stellar Rotational Period [days]
209        210          st_radv            Systemic Radial Velocity [km/s]
215        216               ra                                   RA [deg]
217        218              dec                                  Dec [deg]
233        234          sy_dist                              Distance [pc]
```

In [4]:
```
#redoing the internal index column, 0 to 37
new_column_data=new_column_data.reset_index(drop=True)
print(new_column_data)
```

```
    column_id      abbrev_name                                  full_name
0           1        planet_id                                  Planet ID
1           2          pl_name                                Planet Name
2           3         hostname                                  Host Name
3           4        pl_letter                              Planet Letter
4          10          sy_snum                            Number of Stars
5          11          sy_pnum                         Number of Planets
6          12          sy_mnum                            Number of Moons
7          14  discoverymethod                          Discovery Method
8          15        disc_year                            Discovery Year
9          19    disc_facility                        Discovery Facility
10         33        pl_orbper                      Orbital Period [days]
11         38       pl_orbsmax                 Orbit Semi-Major Axis [au]
12         43        pl_angsep                   Angular Separation [mas]
13         48          pl_rade                Planet Radius [Earth Radius]
14         58        pl_bmasse   Planet Mass or Mass*sin(i) [Earth Mass]
15         69          pl_dens                   Planet Density [g/cm**3]
16         74      pl_orbeccen                                Eccentricity
17         79         pl_insol                Insolation Flux [Earth Flux]
18         84           pl_eqt                Equilibrium Temperature [K]
19         89       pl_orbincl                         Inclination [deg]
20         94       pl_tranmid                     Transit Midpoint [days]
21        106       pl_trandep                         Transit Depth [%]
22        111       pl_trandur                   Transit Duration [hours]
23        142         pl_rvamp            Radial Velocity Amplitude [m/s]
24        157      st_spectype                              Spectral Type
25        164           st_rad              Stellar Radius [Solar Radius]
26        169          st_mass                  Stellar Mass [Solar mass]
27        174           st_met                  Stellar Metallicity [dex]
28        180           st_lum            Stellar Luminosity [log(Solar)]
29        185          st_logg  Stellar Surface Gravity [log10(cm/s**2)]
30        190           st_age                         Stellar Age [Gyr]
31        195          st_dens                  Stellar Density [g/cm**3]
32        200          st_vsin         Stellar Rotational Velocity [km/s]
33        205          st_rotp           Stellar Rotational Period [days]
34        210          st_radv            Systemic Radial Velocity [km/s]
35        216               ra                                   RA [deg]
36        218              dec                                  Dec [deg]
37        234          sy_dist                              Distance [pc]
```

In [5]:
```
#redoing the planet's column id, 1 to length of column, 1 to 38
new_column_data['column_id']=range(1, len(new_column_data) + 1)
print(new_column_data)
```

```
    column_id      abbrev_name                                  full_name
0           1        planet_id                                  Planet ID
1           2          pl_name                                Planet Name
2           3         hostname                                  Host Name
3           4        pl_letter                              Planet Letter
4           5          sy_snum                            Number of Stars
5           6          sy_pnum                         Number of Planets
6           7          sy_mnum                            Number of Moons
7           8  discoverymethod                           Discovery Method
8           9        disc_year                              Discovery Year
9          10    disc_facility                        Discovery Facility
10         11        pl_orbper                      Orbital Period [days]
11         12       pl_orbsmax                 Orbit Semi-Major Axis [au]
12         13        pl_angsep                   Angular Separation [mas]
13         14          pl_rade                Planet Radius [Earth Radius]
14         15        pl_bmasse   Planet Mass or Mass*sin(i) [Earth Mass]
15         16          pl_dens                   Planet Density [g/cm**3]
16         17      pl_orbeccen                               Eccentricity
17         18         pl_insol                Insolation Flux [Earth Flux]
18         19           pl_eqt                Equilibrium Temperature [K]
19         20       pl_orbincl                        Inclination [deg]
20         21       pl_tranmid                    Transit Midpoint [days]
21         22       pl_trandep                        Transit Depth [%]
22         23       pl_trandur                   Transit Duration [hours]
23         24         pl_rvamp            Radial Velocity Amplitude [m/s]
24         25      st_spectype                               Spectral Type
25         26           st_rad              Stellar Radius [Solar Radius]
26         27          st_mass                  Stellar Mass [Solar mass]
27         28           st_met                  Stellar Metallicity [dex]
28         29           st_lum            Stellar Luminosity [log(Solar)]
29         30          st_logg  Stellar Surface Gravity [log10(cm/s**2)]
30         31           st_age                         Stellar Age [Gyr]
31         32          st_dens                  Stellar Density [g/cm**3]
32         33          st_vsin         Stellar Rotational Velocity [km/s]
33         34          st_rotp           Stellar Rotational Period [days]
34         35          st_radv            Systemic Radial Velocity [km/s]
35         36               ra                                   RA [deg]
36         37              dec                                  Dec [deg]
37         38          sy_dist                              Distance [pc]
```

In [6]:
```
#data types and null values for column data
new_column_data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 38 entries, 0 to 37
Data columns (total 3 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   column_id    38 non-null     int64
 1   abbrev_name  38 non-null     str
 2   full_name    38 non-null     str
dtypes: int64(1), str(2)
memory usage: 1.0 KB
```

In [7]:
```
#check for missing values or Na in each column,
#True for missing, false for NOT missing
```

```
row_missing = new_column_data.isna()

#adds the number of rows in each column with missing values
row_total=row_missing.sum()
print(row_total)
```

```
column_id      0
abbrev_name    0
full_name      0
dtype: int64
```

In [8]:
```
#store cleaned data in new CSV file
cleaned_column_data = new_column_data.copy()
cleaned_column_data.to_csv('cleaned_column_data.csv', index=False)
print(cleaned_column_data)
```

```
    column_id      abbrev_name                                 full_name
0           1        planet_id                                 Planet ID
1           2          pl_name                               Planet Name
2           3         hostname                                 Host Name
3           4        pl_letter                             Planet Letter
4           5          sy_snum                           Number of Stars
5           6          sy_pnum                         Number of Planets
6           7          sy_mnum                           Number of Moons
7           8  discoverymethod                         Discovery Method
8           9        disc_year                           Discovery Year
9          10    disc_facility                       Discovery Facility
10         11        pl_orbper                     Orbital Period [days]
11         12       pl_orbsmax                Orbit Semi-Major Axis [au]
12         13        pl_angsep                  Angular Separation [mas]
13         14          pl_rade               Planet Radius [Earth Radius]
14         15        pl_bmasse   Planet Mass or Mass*sin(i) [Earth Mass]
15         16          pl_dens                   Planet Density [g/cm**3]
16         17      pl_orbeccen                              Eccentricity
17         18         pl_insol              Insolation Flux [Earth Flux]
18         19           pl_eqt               Equilibrium Temperature [K]
19         20       pl_orbincl                         Inclination [deg]
20         21       pl_tranmid                    Transit Midpoint [days]
21         22       pl_trandep                         Transit Depth [%]
22         23       pl_trandur                   Transit Duration [hours]
23         24         pl_rvamp           Radial Velocity Amplitude [m/s]
24         25      st_spectype                              Spectral Type
25         26           st_rad             Stellar Radius [Solar Radius]
26         27          st_mass                 Stellar Mass [Solar mass]
27         28           st_met                 Stellar Metallicity [dex]
28         29           st_lum           Stellar Luminosity [log(Solar)]
29         30          st_logg  Stellar Surface Gravity [log10(cm/s**2)]
30         31           st_age                         Stellar Age [Gyr]
31         32          st_dens                 Stellar Density [g/cm**3]
32         33          st_vsin        Stellar Rotational Velocity [km/s]
33         34          st_rotp          Stellar Rotational Period [days]
34         35          st_radv           Systemic Radial Velocity [km/s]
35         36               ra                                  RA [deg]
36         37              dec                                 Dec [deg]
37         38          sy_dist                             Distance [pc]
```

Planet Data

In [9]:
```
#load full planet data file
planet_data=pd.read_csv(r"C:\Users\maybe\OneDrive\Desktop\Databases and Advanced Te


#check first rows to verify it loaded correctly
print(planet_data.head(1))
```

```
   planet_id   pl_name hostname pl_letter    hd_name   hip_name        tic_id  \
0          1  11 Com b   11 Com         b  HD 107383  HIP 60202  TIC 72437047

                   gaia_dr2_id                   gaia_dr3_id  sy_snum  ...  \
0  Gaia DR2 3946945413106333696  Gaia DR3 3946945413106333696        2  ...

   sy_kepmagerr1  sy_kepmagerr2  \
0            NaN            NaN

                                  sy_kepmag_reflink pl_nnotes  st_nphot  \
0  <a refstr=STASSUN_ET_AL__2019 href=https://ui....         2         1

  st_nrvc st_nspec pl_nespec pl_ntranspec pl_ndispec
0       2        0         0            0          0

[1 rows x 320 columns]
```

```
C:\Users\maybe\AppData\Local\Temp\ipykernel_15308\2617141790.py:2: DtypeWarning: Col
umns (0: pl_imppar_reflink, 1: pl_trandep_reflink, 2: pl_trandur_reflink, 3: pl_ratr
or_reflink, 4: pl_occdep_reflink, 5: pl_projobliq_reflink, 6: pl_trueobliq_reflink)
have mixed types. Specify dtype option on import or set low_memory=False.
  planet_data=pd.read_csv(r"C:\Users\maybe\OneDrive\Desktop\Databases and Advanced T
echniques\full_planets_data.csv")
```

In [10]:
```
### data types and null values for planet data
planet_data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 6158 entries, 0 to 6157
Columns: 320 entries, planet_id to pl_ndispec
dtypes: float64(216), int64(24), str(80)
memory usage: 15.0 MB
```

In [11]:
```
#### Find critical columns
critical_planet_columns=['planet_id','pl_name', 'hostname', 'pl_letter', 'sy_snum',
                         'sy_mnum','discoverymethod', 'disc_year', 'disc_facility',
                         'pl_orbper','pl_orbsmax', 'pl_angsep' , 'pl_rade', 'pl_bma
                        'pl_dens', 'pl_orbeccen', 'pl_insol', 'pl_eqt', 'pl_orbincl
                        'pl_trandep', 'pl_trandur', 'pl_rvamp', 'st_spectype', 'st_
                        'st_met', 'st_lum', 'st_logg', 'st_age', 'st_dens', 'st_vsi
                        'st_radv', 'ra', 'dec', 'sy_dist']

new_planet_data = planet_data[critical_planet_columns]
print(new_planet_data)
```

```
      planet_id     pl_name  hostname pl_letter  sy_snum  sy_pnum  sy_mnum  \
0             1    11 Com b    11 Com         b        2        1        0
1             2    11 UMi b    11 UMi         b        1        1        0
2             3    14 And b    14 And         b        1        1        0
3             4    14 Her b    14 Her         b        1        2        0
4             5  16 Cyg B b  16 Cyg B         b        3        1        0
...         ...         ...       ...       ...      ...      ...      ...
6153       6154   ups And b   ups And         b        2        3        0
6154       6155   ups And c   ups And         c        2        3        0
6155       6156   ups And d   ups And         d        2        3        0
6156       6157   ups Leo b   ups Leo         b        1        1        0
6157       6158    xi Aql b    xi Aql         b        1        1        0

      discoverymethod  disc_year                          disc_facility  ...  \
0     Radial Velocity     2007.0                        Xinglong Station  ...
1     Radial Velocity     2009.0  Thueringer Landessternwarte Tautenburg  ...
2     Radial Velocity     2008.0       Okayama Astrophysical Observatory  ...
3     Radial Velocity     2002.0                  W. M. Keck Observatory  ...
4     Radial Velocity     1996.0                  Multiple Observatories  ...
...               ...        ...                                     ...  ...
6153  Radial Velocity     1996.0                        Lick Observatory  ...
6154  Radial Velocity     1999.0                  Multiple Observatories  ...
6155  Radial Velocity     1999.0                  Multiple Observatories  ...
6156  Radial Velocity     2021.0       Okayama Astrophysical Observatory  ...
6157  Radial Velocity     2007.0       Okayama Astrophysical Observatory  ...

       st_lum   st_logg  st_age   st_dens  st_vsin  st_rotp   st_radv  \
0     1.97823  2.450000     NaN       NaN     1.20      NaN  43.36898
1     2.42951  1.930000    1.56       NaN     1.50      NaN -17.52023
2     1.83992  2.550000    4.50       NaN     2.60      NaN -59.72633
3    -0.15273  4.450000    4.60  1.273928     1.00      NaN -13.82260
4     0.09729  4.360000    7.40  1.011029     2.70      NaN -28.10000
...       ...       ...     ...       ...      ...      ...       ...
6153  0.52541  4.125176    5.00  0.369930     9.20     12.0 -28.65500
6154  0.52541  4.125176    5.00  0.369930     9.20     12.0 -28.65500
6155  0.52541  4.125176    5.00  0.369930     9.20     12.0 -28.65500
6156  1.80003  2.460000     NaN       NaN     1.78      NaN       NaN
6157  1.76745  2.610000    7.10       NaN      NaN      NaN -41.66164

              ra        dec   sy_dist
0     185.178779  17.793252   93.1846
1     229.274595  71.823943  125.3210
2     352.824150  39.235837   75.4392
3     242.602101  43.816362   17.9323
4     295.465642  50.516824   21.1397
...          ...        ...       ...
6153   24.198353  41.403815   13.4054
6154   24.198353  41.403815   13.4054
6155   24.198353  41.403815   13.4054
6156  174.237219  -0.823564   52.5973
6157  298.562449   8.461105   56.1858

[6158 rows x 38 columns]
```

In [12]:
```
#check for missing values or Na in each column,
#True for missing, false for NOT missing
```

```
planet_missing = new_planet_data.isna()

#adds the number of rows in each column with missing values
row_total=planet_missing.sum()
print(row_total)
```

```
planet_id             0
pl_name               0
hostname              0
pl_letter             0
sy_snum               0
sy_pnum               0
sy_mnum               0
discoverymethod       0
disc_year             1
disc_facility         0
pl_orbper           337
pl_orbsmax          315
pl_angsep           344
pl_rade              50
pl_bmasse            31
pl_dens             139
pl_orbeccen         938
pl_insol           1842
pl_eqt             1565
pl_orbincl         1522
pl_tranmid         1266
pl_trandep         1777
pl_trandur         1683
pl_rvamp           3667
st_spectype        3847
st_rad              317
st_mass               8
st_met              553
st_lum              311
st_logg             321
st_age             1314
st_dens             586
st_vsin            3881
st_rotp            5268
st_radv            3772
ra                    0
dec                   0
sy_dist              27
dtype: int64
```

In [13]:
```
#we need to drop all planets with missing values under a column, starting with pl_o
required_columns = ['disc_year','pl_orbper','pl_orbsmax', 'pl_angsep' , 'pl_rade',
                        'pl_dens', 'pl_orbeccen', 'pl_insol', 'pl_eqt', 'pl_orbincl
                        'pl_trandep', 'pl_trandur', 'pl_rvamp', 'st_spectype', 'st_
                        'st_met', 'st_lum', 'st_logg', 'st_age', 'st_dens', 'st_vsi
                        'st_radv', 'ra', 'dec', 'sy_dist']

new_planet_data.dropna(subset= required_columns, inplace=True)

#calculating columns with missing values again after dropping planets missing data
```

```
#all are zeros
missing_row_data=new_planet_data.isna()
total_rows=missing_row_data.sum()
print(total_rows)
```

```
planet_id          0
pl_name            0
hostname           0
pl_letter          0
sy_snum            0
sy_pnum            0
sy_mnum            0
discoverymethod    0
disc_year          0
disc_facility      0
pl_orbper          0
pl_orbsmax         0
pl_angsep          0
pl_rade            0
pl_bmasse          0
pl_dens            0
pl_orbeccen        0
pl_insol           0
pl_eqt             0
pl_orbincl         0
pl_tranmid         0
pl_trandep         0
pl_trandur         0
pl_rvamp           0
st_spectype        0
st_rad             0
st_mass            0
st_met             0
st_lum             0
st_logg            0
st_age             0
st_dens            0
st_vsin            0
st_rotp            0
st_radv            0
ra                 0
dec                0
sy_dist            0
dtype: int64
```

In [14]:

```
#calculating rows with missing values again after dropping planets missing data
#all are zeros
missing_column_data = new_planet_data.isna()
total_columns = missing_column_data.sum(axis=1)
#left with 162 entries, or 162 planets that have data in all the required columns
print(total_columns)
```

```
36      0
54      0
55      0
110     0
209     0
       ..
6078    0
6097    0
6100    0
6107    0
6141    0
Length: 162, dtype: int64
```

In [15]:
```
#check to see if rows are full
print(new_planet_data)
```

```
      planet_id     pl_name  hostname pl_letter  sy_snum  sy_pnum  sy_mnum  \
36           37    55 Cnc e    55 Cnc         e        2        7        0
54           55    AU Mic b    AU Mic         b        1        4        0
55           56    AU Mic c    AU Mic         c        1        4        0
110         111  CoRoT-18 b  CoRoT-18         b        1        1        0
209         210    G 9-40 b    G 9-40         b        1        1        0
...         ...         ...       ...       ...      ...      ...      ...
6078       6079   WASP-96 b   WASP-96         b        1        1        0
6097       6098  Wolf 327 b  Wolf 327         b        1        1        0
6100       6101    XO-2 N b    XO-2 N         b        2        3        0
6107       6108      XO-7 b      XO-7         b        1        1        0
6141       6142    pi Men c  HD 39091         c        1        3        0

      discoverymethod  disc_year  \
36    Radial Velocity     2004.0
54            Transit     2020.0
55            Transit     2021.0
110           Transit     2011.0
209           Transit     2019.0
...               ...        ...
6078          Transit     2014.0
6097          Transit     2024.0
6100          Transit     2007.0
6107          Transit     2019.0
6141          Transit     2018.0

                                     disc_facility  ...   st_lum  st_logg  \
36                            McDonald Observatory  ... -0.19695    4.430
54    Transiting Exoplanet Survey Satellite (TESS)  ... -0.97757    4.370
55    Transiting Exoplanet Survey Satellite (TESS)  ... -0.97757    4.370
110                                          CoRoT  ... -0.17000    4.400
209                                             K2  ... -1.96019    4.840
...                                            ...  ...      ...      ...
6078                                      SuperWASP  ...  0.03226    4.420
6097  Transiting Exoplanet Survey Satellite (TESS)  ... -1.62986    4.870
6100                                             XO  ... -0.08823    4.433
6107                                             XO  ...  0.43132    4.246
6141  Transiting Exoplanet Survey Satellite (TESS)  ...  0.15836    4.430

      st_age   st_dens  st_vsin  st_rotp   st_radv          ra        dec  \
36     5.500  1.521203     1.23    38.80  27.40949  133.146837  28.329815
54     0.019  1.750000     9.20     4.86  -6.31031  311.291137 -31.342450
55     0.019  1.750000     9.20     4.86  -6.31031  311.291137 -31.342450
110    0.600  1.350000     8.00     5.40  29.55000   98.172417  -0.031605
209    9.900  1.620000     2.00    30.00  14.65000  134.718851  21.074796
...      ...       ...      ...      ...       ...         ...        ...
6078   8.000  1.299000     1.50    20.16  -1.23300    1.046570 -47.360635
6097   4.100  8.540000     2.00    44.40   9.39000  148.378320  35.570548
6100   7.800  1.410000     1.07    28.60  46.93290  117.026769  50.225147
6107   1.180  0.451084     6.00     3.62 -12.98300  277.478075  85.233321
6141   2.980  0.944584     3.16    18.30  10.70916   84.299280 -80.464604

       sy_dist
36     12.5855
54      9.7221
55      9.7221
```

```
110   764.8890
209    27.9276
...        ...
6078  352.4640
6097   28.5321
6100  154.2730
6107  234.1490
6141   18.2702

[162 rows x 38 columns]
```

In [16]:
```
#redoing the internal index column, 0 to 70
new_planet_data=new_planet_data.reset_index(drop=True)
print(new_planet_data)
```

```
     planet_id     pl_name  hostname pl_letter  sy_snum  sy_pnum  sy_mnum  \
0           37    55 Cnc e    55 Cnc         e        2        7        0
1           55    AU Mic b    AU Mic         b        1        4        0
2           56    AU Mic c    AU Mic         c        1        4        0
3          111  CoRoT-18 b  CoRoT-18         b        1        1        0
4          210    G 9-40 b    G 9-40         b        1        1        0
..         ...         ...       ...       ...      ...      ...      ...
157       6079   WASP-96 b   WASP-96         b        1        1        0
158       6098  Wolf 327 b  Wolf 327         b        1        1        0
159       6101    XO-2 N b    XO-2 N         b        2        3        0
160       6108      XO-7 b      XO-7         b        1        1        0
161       6142    pi Men c  HD 39091         c        1        3        0

     discoverymethod  disc_year                                disc_facility  \
0    Radial Velocity     2004.0                           McDonald Observatory
1            Transit     2020.0  Transiting Exoplanet Survey Satellite (TESS)
2            Transit     2021.0  Transiting Exoplanet Survey Satellite (TESS)
3            Transit     2011.0                                          CoRoT
4            Transit     2019.0                                             K2
..               ...        ...                                            ...
157          Transit     2014.0                                      SuperWASP
158          Transit     2024.0  Transiting Exoplanet Survey Satellite (TESS)
159          Transit     2007.0                                             XO
160          Transit     2019.0                                             XO
161          Transit     2018.0  Transiting Exoplanet Survey Satellite (TESS)

     ...   st_lum  st_logg  st_age   st_dens  st_vsin  st_rotp   st_radv  \
0    ... -0.19695    4.430   5.500  1.521203     1.23    38.80  27.40949
1    ... -0.97757    4.370   0.019  1.750000     9.20     4.86  -6.31031
2    ... -0.97757    4.370   0.019  1.750000     9.20     4.86  -6.31031
3    ... -0.17000    4.400   0.600  1.350000     8.00     5.40  29.55000
4    ... -1.96019    4.840   9.900  1.620000     2.00    30.00  14.65000
..   ...      ...      ...     ...       ...      ...      ...       ...
157  ...  0.03226    4.420   8.000  1.299000     1.50    20.16  -1.23300
158  ... -1.62986    4.870   4.100  8.540000     2.00    44.40   9.39000
159  ... -0.08823    4.433   7.800  1.410000     1.07    28.60  46.93290
160  ...  0.43132    4.246   1.180  0.451084     6.00     3.62 -12.98300
161  ...  0.15836    4.430   2.980  0.944584     3.16    18.30  10.70916

             ra        dec   sy_dist
0    133.146837  28.329815   12.5855
1    311.291137 -31.342450    9.7221
2    311.291137 -31.342450    9.7221
3     98.172417  -0.031605  764.8890
4    134.718851  21.074796   27.9276
..          ...        ...       ...
157    1.046570 -47.360635  352.4640
158  148.378320  35.570548   28.5321
159  117.026769  50.225147  154.2730
160  277.478075  85.233321  234.1490
161   84.299280 -80.464604   18.2702

[162 rows x 38 columns]
```

In [17]:
```
#renaming the planet's id with new table, 1 to length of column, 1 to 71
new_planet_data['planet_id']=range(1, len(new_planet_data) + 1)
```

```
print(new_planet_data)
```

```
     planet_id     pl_name  hostname pl_letter  sy_snum  sy_pnum  sy_mnum  \
0            1    55 Cnc e    55 Cnc         e        2        7        0
1            2    AU Mic b    AU Mic         b        1        4        0
2            3    AU Mic c    AU Mic         c        1        4        0
3            4  CoRoT-18 b  CoRoT-18         b        1        1        0
4            5    G 9-40 b    G 9-40         b        1        1        0
..         ...         ...       ...       ...      ...      ...      ...
157        158   WASP-96 b   WASP-96         b        1        1        0
158        159  Wolf 327 b  Wolf 327         b        1        1        0
159        160    XO-2 N b    XO-2 N         b        2        3        0
160        161      XO-7 b      XO-7         b        1        1        0
161        162    pi Men c  HD 39091         c        1        3        0

     discoverymethod  disc_year                                disc_facility  \
0    Radial Velocity     2004.0                           McDonald Observatory
1            Transit     2020.0  Transiting Exoplanet Survey Satellite (TESS)
2            Transit     2021.0  Transiting Exoplanet Survey Satellite (TESS)
3            Transit     2011.0                                          CoRoT
4            Transit     2019.0                                             K2
..               ...        ...                                            ...
157          Transit     2014.0                                       SuperWASP
158          Transit     2024.0  Transiting Exoplanet Survey Satellite (TESS)
159          Transit     2007.0                                             XO
160          Transit     2019.0                                             XO
161          Transit     2018.0  Transiting Exoplanet Survey Satellite (TESS)

     ...   st_lum  st_logg  st_age   st_dens  st_vsin  st_rotp   st_radv  \
0    ... -0.19695    4.430   5.500  1.521203     1.23    38.80  27.40949
1    ... -0.97757    4.370   0.019  1.750000     9.20     4.86  -6.31031
2    ... -0.97757    4.370   0.019  1.750000     9.20     4.86  -6.31031
3    ... -0.17000    4.400   0.600  1.350000     8.00     5.40  29.55000
4    ... -1.96019    4.840   9.900  1.620000     2.00    30.00  14.65000
..   ...      ...      ...     ...       ...      ...      ...       ...
157  ...  0.03226    4.420   8.000  1.299000     1.50    20.16  -1.23300
158  ... -1.62986    4.870   4.100  8.540000     2.00    44.40   9.39000
159  ... -0.08823    4.433   7.800  1.410000     1.07    28.60  46.93290
160  ...  0.43132    4.246   1.180  0.451084     6.00     3.62 -12.98300
161  ...  0.15836    4.430   2.980  0.944584     3.16    18.30  10.70916

             ra        dec   sy_dist
0    133.146837  28.329815   12.5855
1    311.291137 -31.342450    9.7221
2    311.291137 -31.342450    9.7221
3     98.172417  -0.031605  764.8890
4    134.718851  21.074796   27.9276
..          ...        ...       ...
157    1.046570 -47.360635  352.4640
158  148.378320  35.570548   28.5321
159  117.026769  50.225147  154.2730
160  277.478075  85.233321  234.1490
161   84.299280 -80.464604   18.2702

[162 rows x 38 columns]
```

Stage 2. Model your data

In [18]:
```
## instead of having two datasets, make the columns names into their full names for

#rename columns here
new_planet_data = new_planet_data.rename(columns={'planet_id': 'Planet ID', 'pl_nam
                                                  'hostname': 'Host Name', 'pl_lett
                                                  'sy_snum':'Number of Stars', 'sy_
                                                  'sy_mnum':'Number of Moons','disc
                                                  'disc_year':'Discovery Year','dis
                                                  'pl_orbper':'Orbital Period [days
                                                  'pl_angsep':'Angular Separation [
                                                  'pl_bmasse':'Planet Mass or Mass*
                                                  'pl_orbeccen': 'Eccentricity', 'p
                                                  'pl_eqt': 'Equilibrium Temperatur
                                                  'pl_tranmid':'Transit Midpoint [d
                                                  'pl_trandur':'Transit Duration [h
                                                  'st_spectype':'Spectral Type', 's
                                                  'st_mass':'Stellar Mass [Solar ma
                                                  'st_lum':'Stellar Luminosity [log
                                                  'st_age':'Stellar Age [Gyr]', 'st
                                                  'st_vsin':'Stellar Rotational Vel
                                                  'st_radv':'Systemic Radial Veloci
                                                  'sy_dist':'Distance [pc]'})
print(new_planet_data)
```

```
     Planet ID Planet Name Host Name Planet Letter  Number of Stars  \
0            1    55 Cnc e    55 Cnc             e                2
1            2    AU Mic b    AU Mic             b                1
2            3    AU Mic c    AU Mic             c                1
3            4  CoRoT-18 b  CoRoT-18             b                1
4            5    G 9-40 b    G 9-40             b                1
..         ...         ...       ...           ...              ...
157        158   WASP-96 b   WASP-96             b                1
158        159  Wolf 327 b  Wolf 327             b                1
159        160    XO-2 N b    XO-2 N             b                2
160        161      XO-7 b      XO-7             b                1
161        162    pi Men c  HD 39091             c                1

     Number of Planets  Number of Moons Discovery Method  Discovery Year  \
0                    7                0  Radial Velocity          2004.0
1                    4                0          Transit          2020.0
2                    4                0          Transit          2021.0
3                    1                0          Transit          2011.0
4                    1                0          Transit          2019.0
..                 ...              ...              ...             ...
157                  1                0          Transit          2014.0
158                  1                0          Transit          2024.0
159                  3                0          Transit          2007.0
160                  1                0          Transit          2019.0
161                  3                0          Transit          2018.0

                              Discovery Facility  ...  \
0                          McDonald Observatory  ...
1    Transiting Exoplanet Survey Satellite (TESS)  ...
2    Transiting Exoplanet Survey Satellite (TESS)  ...
3                                          CoRoT  ...
4                                             K2  ...
..                                           ...  ...
157                                    SuperWASP  ...
158  Transiting Exoplanet Survey Satellite (TESS)  ...
159                                           XO  ...
160                                           XO  ...
161  Transiting Exoplanet Survey Satellite (TESS)  ...

     Stellar Luminosity [log(Solar)]  \
0                          -0.19695
1                          -0.97757
2                          -0.97757
3                          -0.17000
4                          -1.96019
..                              ...
157                         0.03226
158                        -1.62986
159                        -0.08823
160                         0.43132
161                         0.15836

     Stellar Surface Gravity [log10(cm/s**2)]  Stellar Age [Gyr]  \
0                                       4.430              5.500
1                                       4.370              0.019
2                                       4.370              0.019
```

```
3                                       4.400              0.600
4                                       4.840              9.900
..                                        ...                ...
157                                     4.420              8.000
158                                     4.870              4.100
159                                     4.433              7.800
160                                     4.246              1.180
161                                     4.430              2.980

     Stellar Density [g/cm**3]  Stellar Rotational Velocity [km/s]  \
0                     1.521203                                1.23
1                     1.750000                                9.20
2                     1.750000                                9.20
3                     1.350000                                8.00
4                     1.620000                                2.00
..                         ...                                 ...
157                   1.299000                                1.50
158                   8.540000                                2.00
159                   1.410000                                1.07
160                   0.451084                                6.00
161                   0.944584                                3.16

     Stellar Rotational Period [days]  Systemic Radial Velocity [km/s]  \
0                               38.80                        27.40949
1                                4.86                        -6.31031
2                                4.86                        -6.31031
3                                5.40                        29.55000
4                               30.00                        14.65000
..                                ...                             ...
157                             20.16                        -1.23300
158                             44.40                         9.39000
159                             28.60                        46.93290
160                              3.62                       -12.98300
161                             18.30                        10.70916

       RA [deg]  Dec [deg]  Distance [pc]
0    133.146837  28.329815        12.5855
1    311.291137 -31.342450         9.7221
2    311.291137 -31.342450         9.7221
3     98.172417  -0.031605       764.8890
4    134.718851  21.074796        27.9276
..          ...        ...            ...
157    1.046570 -47.360635       352.4640
158  148.378320  35.570548        28.5321
159  117.026769  50.225147       154.2730
160  277.478075  85.233321       234.1490
161   84.299280 -80.464604        18.2702

[162 rows x 38 columns]
```

In [19]:

```
##add new columns to dataset, Star ID, Discovery ID, Solar System ID

#variables for number of columns and rows or current data
number_columns = len(new_planet_data.columns)
number_rows = len(new_planet_data)
```

```
## add new columns using data.insert(location, column name, rows) at the end of the
new_planet_data.insert(number_columns, 'Star ID', range(1,  number_rows + 1))

new_planet_data.insert(number_columns + 1, 'System ID', range(1, number_rows + 1))

new_planet_data.insert(number_columns + 2, 'Discovery ID', range(1, number_rows + 1

print(new_planet_data)
```

```
     Planet ID Planet Name Host Name Planet Letter  Number of Stars  \
0            1    55 Cnc e    55 Cnc             e                2
1            2    AU Mic b    AU Mic             b                1
2            3    AU Mic c    AU Mic             c                1
3            4  CoRoT-18 b  CoRoT-18             b                1
4            5    G 9-40 b    G 9-40             b                1
..         ...         ...       ...           ...              ...
157        158   WASP-96 b   WASP-96             b                1
158        159  Wolf 327 b  Wolf 327             b                1
159        160    XO-2 N b    XO-2 N             b                2
160        161      XO-7 b      XO-7             b                1
161        162    pi Men c  HD 39091             c                1

     Number of Planets  Number of Moons Discovery Method  Discovery Year  \
0                    7                0  Radial Velocity          2004.0
1                    4                0          Transit          2020.0
2                    4                0          Transit          2021.0
3                    1                0          Transit          2011.0
4                    1                0          Transit          2019.0
..                 ...              ...              ...             ...
157                  1                0          Transit          2014.0
158                  1                0          Transit          2024.0
159                  3                0          Transit          2007.0
160                  1                0          Transit          2019.0
161                  3                0          Transit          2018.0

                              Discovery Facility  ...  \
0                          McDonald Observatory  ...
1    Transiting Exoplanet Survey Satellite (TESS)  ...
2    Transiting Exoplanet Survey Satellite (TESS)  ...
3                                          CoRoT  ...
4                                             K2  ...
..                                           ...  ...
157                                     SuperWASP  ...
158  Transiting Exoplanet Survey Satellite (TESS)  ...
159                                           XO  ...
160                                           XO  ...
161  Transiting Exoplanet Survey Satellite (TESS)  ...

     Stellar Density [g/cm**3]  Stellar Rotational Velocity [km/s]  \
0                     1.521203                                1.23
1                     1.750000                                9.20
2                     1.750000                                9.20
3                     1.350000                                8.00
4                     1.620000                                2.00
..                         ...                                 ...
157                   1.299000                                1.50
158                   8.540000                                2.00
159                   1.410000                                1.07
160                   0.451084                                6.00
161                   0.944584                                3.16

     Stellar Rotational Period [days]  Systemic Radial Velocity [km/s]  \
0                               38.80                          27.40949
1                                4.86                          -6.31031
2                                4.86                          -6.31031
```

```
3                                5.40                          29.55000
4                               30.00                          14.65000
..                                ...                               ...
157                             20.16                          -1.23300
158                             44.40                           9.39000
159                             28.60                          46.93290
160                              3.62                         -12.98300
161                             18.30                          10.70916

       RA [deg]  Dec [deg]  Distance [pc]  Star ID  System ID  Discovery ID
0    133.146837  28.329815        12.5855        1          1             1
1    311.291137 -31.342450         9.7221        2          2             2
2    311.291137 -31.342450         9.7221        3          3             3
3     98.172417  -0.031605       764.8890        4          4             4
4    134.718851  21.074796        27.9276        5          5             5
..          ...        ...            ...      ...        ...           ...
157    1.046570 -47.360635       352.4640      158        158           158
158  148.378320  35.570548        28.5321      159        159           159
159  117.026769  50.225147       154.2730      160        160           160
160  277.478075  85.233321       234.1490      161        161           161
161   84.299280 -80.464604        18.2702      162        162           162

[162 rows x 41 columns]
```

In [20]:

```
## Create a new dataset for Stars, Solar System, and Discoveries

stars_data = new_planet_data['Star ID']
print(stars_data)

system_data =  new_planet_data['System ID']
print(system_data)

discoveries_data =  new_planet_data['Discovery ID']
print(discoveries_data)
```

```
0        1
1        2
2        3
3        4
4        5
      ...
157    158
158    159
159    160
160    161
161    162
Name: Star ID, Length: 162, dtype: int64
0        1
1        2
2        3
3        4
4        5
      ...
157    158
158    159
159    160
160    161
161    162
Name: System ID, Length: 162, dtype: int64
0        1
1        2
2        3
3        4
4        5
      ...
157    158
158    159
159    160
160    161
161    162
Name: Discovery ID, Length: 162, dtype: int64
```

In [21]:

```
## Add data to new data sets
#i need to find the columns, then i need add the columns to the new dataset

#Stars
#Spectral Type, Stellar Radius [Solar Radius],
#Stellar Mass [Solar mass], Stellar Metallicity [dex],
#Stellar Luminosity [log(Solar)], Stellar Surface Gravity [log10(cm/s**2)],
#Stellar Age [Gyr], Stellar Density [g/cm**3], Stellar Rotational Velocity [km/s]
#Stellar Rotational Period [days], Systemic Radial Velocity [km/s],

#find columns and assign to variable
stars_columns=['Star ID','Spectral Type', 'Stellar Radius [Solar Radius]',
'Stellar Mass [Solar mass]', 'Stellar Metallicity [dex]',
'Stellar Luminosity [log(Solar)]', 'Stellar Surface Gravity [log10(cm/s**2)]',
'Stellar Age [Gyr]', 'Stellar Density [g/cm**3]', 'Stellar Rotational Velocity [km/
'Stellar Rotational Period [days]', 'Systemic Radial Velocity [km/s]']

#redefine stars_data using new varable with column names
```

```
stars_data = new_planet_data[stars_columns]
print(stars_data)
```

```
     Star ID Spectral Type  Stellar Radius [Solar Radius]  \
0          1          G8 V                         0.9430
1          2          M1 V                         0.8620
2          3          M1 V                         0.8620
3          4          G9 V                         1.0000
4          5        M2.5 V                         0.3026
..       ...           ...                            ...
157      158            G8                         1.0500
158      159        M2.5 V                         0.4060
159      160          K0 V                         0.9900
160      161          G0 V                         1.4800
161      162          G0 V                         1.1700

     Stellar Mass [Solar mass]  Stellar Metallicity [dex]  \
0                       0.9050                      0.350
1                       0.6350                      0.010
2                       0.6350                      0.010
3                       0.9500                     -0.100
4                       0.2952                     -0.070
..                         ...                        ...
157                     1.0600                      0.140
158                     0.4050                     -0.170
159                     0.9710                      0.430
160                     1.4050                      0.432
161                     1.0700                      0.090

     Stellar Luminosity [log(Solar)]  \
0                          -0.19695
1                          -0.97757
2                          -0.97757
3                          -0.17000
4                          -1.96019
..                              ...
157                         0.03226
158                        -1.62986
159                        -0.08823
160                         0.43132
161                         0.15836

     Stellar Surface Gravity [log10(cm/s**2)]  Stellar Age [Gyr]  \
0                                       4.430             5.500
1                                       4.370             0.019
2                                       4.370             0.019
3                                       4.400             0.600
4                                       4.840             9.900
..                                        ...               ...
157                                     4.420             8.000
158                                     4.870             4.100
159                                     4.433             7.800
160                                     4.246             1.180
161                                     4.430             2.980

     Stellar Density [g/cm**3]  Stellar Rotational Velocity [km/s]  \
0                     1.521203                                1.23
1                     1.750000                                9.20
2                     1.750000                                9.20
```

```
3                    1.350000                                8.00
4                    1.620000                                2.00
..                        ...                                 ...
157                  1.299000                                1.50
158                  8.540000                                2.00
159                  1.410000                                1.07
160                  0.451084                                6.00
161                  0.944584                                3.16

     Stellar Rotational Period [days]  Systemic Radial Velocity [km/s]
0                               38.80                        27.40949
1                                4.86                        -6.31031
2                                4.86                        -6.31031
3                                5.40                        29.55000
4                               30.00                        14.65000
..                                ...                             ...
157                             20.16                        -1.23300
158                             44.40                         9.39000
159                             28.60                        46.93290
160                              3.62                       -12.98300
161                             18.30                        10.70916

[162 rows x 12 columns]
```

In [22]:

```
## there is a foreign key in stars data that links it to solar system
# we need to add system_id at the end of the dataset like we need for exoplanets ab

#variables for number of columns and rows or current data
number_columns = len(stars_data.columns)

number_rows = len(stars_data)

## add new columns using data.insert(location, column name, rows)
stars_data.insert(number_columns, 'System ID', range(1, number_rows + 1))

print(stars_data)
```

```
12
162
     Star ID Spectral Type  Stellar Radius [Solar Radius]  \
0          1          G8 V                         0.9430
1          2          M1 V                         0.8620
2          3          M1 V                         0.8620
3          4          G9 V                         1.0000
4          5        M2.5 V                         0.3026
..       ...           ...                            ...
157      158            G8                         1.0500
158      159        M2.5 V                         0.4060
159      160          K0 V                         0.9900
160      161          G0 V                         1.4800
161      162          G0 V                         1.1700

     Stellar Mass [Solar mass]  Stellar Metallicity [dex]  \
0                       0.9050                      0.350
1                       0.6350                      0.010
2                       0.6350                      0.010
3                       0.9500                     -0.100
4                       0.2952                     -0.070
..                         ...                        ...
157                     1.0600                      0.140
158                     0.4050                     -0.170
159                     0.9710                      0.430
160                     1.4050                      0.432
161                     1.0700                      0.090

     Stellar Luminosity [log(Solar)]  \
0                          -0.19695
1                          -0.97757
2                          -0.97757
3                          -0.17000
4                          -1.96019
..                              ...
157                         0.03226
158                        -1.62986
159                        -0.08823
160                         0.43132
161                         0.15836

     Stellar Surface Gravity [log10(cm/s**2)]  Stellar Age [Gyr]  \
0                                       4.430             5.500
1                                       4.370             0.019
2                                       4.370             0.019
3                                       4.400             0.600
4                                       4.840             9.900
..                                        ...               ...
157                                     4.420             8.000
158                                     4.870             4.100
159                                     4.433             7.800
160                                     4.246             1.180
161                                     4.430             2.980

     Stellar Density [g/cm**3]  Stellar Rotational Velocity [km/s]  \
0                     1.521203                                1.23
```

```
1                    1.750000                                9.20
2                    1.750000                                9.20
3                    1.350000                                8.00
4                    1.620000                                2.00
..                        ...                                 ...
157                  1.299000                                1.50
158                  8.540000                                2.00
159                  1.410000                                1.07
160                  0.451084                                6.00
161                  0.944584                                3.16

     Stellar Rotational Period [days]  Systemic Radial Velocity [km/s]  \
0                               38.80                         27.40949
1                                4.86                         -6.31031
2                                4.86                         -6.31031
3                                5.40                         29.55000
4                               30.00                         14.65000
..                                ...                              ...
157                             20.16                         -1.23300
158                             44.40                          9.39000
159                             28.60                         46.93290
160                              3.62                        -12.98300
161                             18.30                         10.70916

     System ID
0            1
1            2
2            3
3            4
4            5
..         ...
157        158
158        159
159        160
160        161
161        162

[162 rows x 13 columns]
```

In [23]:

```
#System
#System ID, Host Name, Number of Stars, Number of Planets, RA [deg],
#Dec [deg], Distance [pc]

system_columns=['System ID', 'Host Name', 'Number of Stars', 'Number of Planets', '
'Dec [deg]', 'Distance [pc]']

system_data = new_planet_data[system_columns]
print(system_data)


#Discoveries
#Discovery ID, Discovery Method, Discovery Year, Discovery Facility

discovery_columns=['Discovery ID', 'Discovery Method', 'Discovery Year', 'Discovery
```

```
discoveries_data = new_planet_data[discovery_columns]
print(discoveries_data)
```

```
     System ID Host Name  Number of Stars  Number of Planets    RA [deg]  \
0            1    55 Cnc                2                  7  133.146837
1            2    AU Mic                1                  4  311.291137
2            3    AU Mic                1                  4  311.291137
3            4  CoRoT-18                1                  1   98.172417
4            5    G 9-40                1                  1  134.718851
..         ...       ...              ...                ...         ...
157        158   WASP-96                1                  1    1.046570
158        159  Wolf 327                1                  1  148.378320
159        160    XO-2 N                2                  3  117.026769
160        161      XO-7                1                  1  277.478075
161        162  HD 39091                1                  3   84.299280

     Dec [deg]  Distance [pc]
0    28.329815        12.5855
1   -31.342450         9.7221
2   -31.342450         9.7221
3    -0.031605       764.8890
4    21.074796        27.9276
..         ...            ...
157 -47.360635       352.4640
158  35.570548        28.5321
159  50.225147       154.2730
160  85.233321       234.1490
161 -80.464604        18.2702

[162 rows x 7 columns]
     Discovery ID Discovery Method  Discovery Year  \
0               1  Radial Velocity          2004.0
1               2          Transit          2020.0
2               3          Transit          2021.0
3               4          Transit          2011.0
4               5          Transit          2019.0
..            ...              ...             ...
157           158          Transit          2014.0
158           159          Transit          2024.0
159           160          Transit          2007.0
160           161          Transit          2019.0
161           162          Transit          2018.0

                              Discovery Facility
0                          McDonald Observatory
1    Transiting Exoplanet Survey Satellite (TESS)
2    Transiting Exoplanet Survey Satellite (TESS)
3                                          CoRoT
4                                             K2
..                                           ...
157                                    SuperWASP
158  Transiting Exoplanet Survey Satellite (TESS)
159                                           XO
160                                           XO
161  Transiting Exoplanet Survey Satellite (TESS)

[162 rows x 4 columns]
```

In [24]:
```
#Planet ID, Planet Name, Planet Letter, Number of Moons, Orbital Period [days], Orb
#Angular Separation [mas], Planet Radius [Earth Radius], Planet Mass or Mass*sin(i)
#Eccentricity, Insolation Flux [Earth Flux], Equilibrium Temperature [K], Inclinati
#Transit Depth [%], Transit Duration [hours], Radial Velocity Amplitude [m/s], Star

planet_columns=['Planet ID', 'Planet Name', 'Planet Letter', 'Number of Moons', 'Or
'Angular Separation [mas]', 'Planet Radius [Earth Radius]', 'Planet Mass or Mass*si
'Eccentricity', 'Insolation Flux [Earth Flux]', 'Equilibrium Temperature [K]', 'Inc
'Transit Depth [%]', 'Transit Duration [hours]', 'Radial Velocity Amplitude [m/s]',

new_planet_data = new_planet_data[planet_columns]
print(new_planet_data)
```

```
     Planet ID Planet Name Planet Letter  Number of Moons  \
0            1    55 Cnc e             e                0
1            2    AU Mic b             b                0
2            3    AU Mic c             c                0
3            4  CoRoT-18 b             b                0
4            5    G 9-40 b             b                0
..         ...         ...           ...              ...
157        158   WASP-96 b             b                0
158        159  Wolf 327 b             b                0
159        160    XO-2 N b             b                0
160        161      XO-7 b             b                0
161        162    pi Men c             c                0

     Orbital Period [days]  Orbit Semi-Major Axis [au]  \
0                 0.736547                     0.01544
1                 8.463446                     0.07000
2                18.859023                     0.11900
3                 1.900069                     0.02950
4                 5.745998                     0.04180
..                     ...                         ...
157               3.425260                     0.04530
158               0.573474                     0.01000
159               2.615862                     0.03680
160               2.864142                     0.04421
161               6.267840                     0.06839

     Angular Separation [mas]  Planet Radius [Earth Radius]  \
0                      1.2300                      1.875000
1                      7.2000                      4.790000
2                     12.2000                      2.790000
3                      0.0386                     14.680000
4                      1.5000                      1.900000
..                        ...                           ...
157                    0.1290                     13.450000
158                    0.3500                      1.240000
159                    0.2390                     11.130537
160                    0.1890                     15.389957
161                    3.7400                      2.018900

     Planet Mass or Mass*sin(i) [Earth Mass]  Planet Density [g/cm**3]  ...  \
0                                    7.99000                     6.660  ...
1                                    8.99000                     0.490  ...
2                                   14.46000                     3.900  ...
3                                 1102.82000                     2.200  ...
4                                    4.00000                     3.200  ...
..                                       ...                       ...  ...
157                                152.55000                     0.370  ...
158                                  2.53000                     7.240  ...
159                                179.89178                     0.715  ...
160                                225.34147                     0.340  ...
161                                  3.63000                     2.800  ...

     Insolation Flux [Earth Flux]  Equilibrium Temperature [K]  \
0                       2657.8322                       1958.0
1                         21.2000                        554.8
2                          7.3000                        424.7
```

```
3                        823.7009                       1550.0
4                          6.2700                        440.6
..                            ...                          ...
157                      517.0000                       1285.0
158                      233.9000                        996.0
159                      428.0360                       1328.0
160                     1540.0000                       1743.0
161                      309.0000                       1169.8

     Inclination [deg]  Transit Midpoint [days]  Transit Depth [%]  \
0               83.590              2457063.210           0.038467
1               88.390              2458330.351           0.249000
2               89.460              2458342.223           0.085000
3               86.500              2455321.724           2.571701
4               89.030              2459503.327           0.365000
..                 ...                      ...                ...
157             85.600              2456258.062           1.380000
158             84.890              2459252.981           0.081839
159             88.010              2454508.738           1.336300
160             83.450              2457917.475           0.909000
161             87.045              2458425.789           0.026759

     Transit Duration [hours]  Radial Velocity Amplitude [m/s]  Star ID  \
0                    1.543900                            6.020        1
1                    3.240000                            3.800        2
2                    4.290000                            4.696        3
3                    2.387000                          590.000        4
4                    1.693000                            3.220        5
..                        ...                              ...      ...
157                  2.426400                           62.000      158
158                  0.900451                            3.540      159
159                  2.683900                           90.170      160
160                  2.772000                           80.500      161
161                  2.952024                            1.185      162

     System ID  Discovery ID
0            1             1
1            2             2
2            3             3
3            4             4
4            5             5
..         ...           ...
157        158           158
158        159           159
159        160           160
160        161           161
161        162           162

[162 rows x 21 columns]
```

In [25]:

```
#count of entries before drop_duplicates
star_entries_before = len(stars_data)
print( 'stars before: ', star_entries_before)

stars_data.drop_duplicates(subset=['Spectral Type', 'Stellar Radius [Solar Radius]'
'Stellar Mass [Solar mass]', 'Stellar Metallicity [dex]',
```

```
'Stellar Luminosity [log(Solar)]', 'Stellar Surface Gravity [log10(cm/s**2)]',
'Stellar Age [Gyr]', 'Stellar Density [g/cm**3]', 'Stellar Rotational Velocity [km/
'Stellar Rotational Period [days]', 'Systemic Radial Velocity [km/s]'], inplace=Tru

#after drops
star_entries_after = len(stars_data)
print( 'stars after:', star_entries_after)

#difference
star_entries_dropped = star_entries_before - star_entries_after
print("Duplicates Removed:", star_entries_dropped)
```

```
stars before:  162
stars after: 131
Duplicates Removed: 31
```

In [26]:
```
#count of entries before drop_duplicates
system_entries_before = len(system_data)
print('system before: ', system_entries_before)

system_data.drop_duplicates(subset=['Host Name'], inplace=True)

#after
system_entries_after = len(system_data)
print( 'system after:', system_entries_after)

#difference
system_entries_dropped = system_entries_before - system_entries_after

print("Duplicates Removed:", system_entries_dropped)
```

```
system before:  162
system after: 130
Duplicates Removed: 32
```

In [27]:
```
#count of entries before drop_duplicates
discovery_entries_before = len(discoveries_data)
print('discoveries before: ', discovery_entries_before)

discoveries_data.drop_duplicates(subset=['Discovery Method', 'Discovery Year', 'Dis

#after
discovery_entries_after = len(discoveries_data)
print( 'discoveries after:', discovery_entries_after)

#difference
discovery_entries_dropped = discovery_entries_before - discovery_entries_after

print("Duplicates Removed:", discovery_entries_dropped)
```

```
discoveries before:  162
discoveries after: 56
Duplicates Removed: 106
```

In [28]:
```
#count of entries before drop_duplicates
planet_entries_before = len(new_planet_data)
print('planets before: ', planet_entries_before)
```

```
new_planet_data.drop_duplicates(subset=['Planet Name'], inplace=True)

#after
planet_entries_after = len(new_planet_data)
print( 'planet after:', planet_entries_after)

#difference
planet_entries_dropped = planet_entries_before - planet_entries_after

print("Duplicates Removed:", planet_entries_dropped)
```

```
planets before:  162
planet after: 162
Duplicates Removed: 0
```

In [29]:

```
#check for missing values or Na in each column,
#True for missing, false for NOT missing

missing_values = new_planet_data.isna()

#adds the number of rows in each column with missing values
row_total=missing_values.sum()
print(row_total)
print('_________________________Exoplanets_________________________')

missing_values = stars_data.isna()

#adds the number of rows in each column with missing values
row_total=missing_values.sum()
print(row_total)
print('_________________________Stars_____________________________')


missing_values = system_data.isna()

row_total=missing_values.sum()
print(row_total)
print('_______________________Solar System_________________________')


missing_values = discoveries_data.isna()

row_total=missing_values.sum()
print(row_total)
print('_______________________Discoveries__________________________')
```

```
Planet ID                                   0
Planet Name                                 0
Planet Letter                               0
Number of Moons                             0
Orbital Period [days]                       0
Orbit Semi-Major Axis [au]                  0
Angular Separation [mas]                    0
Planet Radius [Earth Radius]                0
Planet Mass or Mass*sin(i) [Earth Mass]     0
Planet Density [g/cm**3]                    0
Eccentricity                                0
Insolation Flux [Earth Flux]                0
Equilibrium Temperature [K]                 0
Inclination [deg]                           0
Transit Midpoint [days]                     0
Transit Depth [%]                           0
Transit Duration [hours]                    0
Radial Velocity Amplitude [m/s]             0
Star ID                                     0
System ID                                   0
Discovery ID                                0
dtype: int64
_________________________Exoplanets_________________________
Star ID                                      0
Spectral Type                                0
Stellar Radius [Solar Radius]                0
Stellar Mass [Solar mass]                    0
Stellar Metallicity [dex]                    0
Stellar Luminosity [log(Solar)]              0
Stellar Surface Gravity [log10(cm/s**2)]     0
Stellar Age [Gyr]                            0
Stellar Density [g/cm**3]                    0
Stellar Rotational Velocity [km/s]           0
Stellar Rotational Period [days]             0
Systemic Radial Velocity [km/s]              0
System ID                                    0
dtype: int64
_________________________Stars_______________________________
System ID            0
Host Name            0
Number of Stars      0
Number of Planets    0
RA [deg]             0
Dec [deg]            0
Distance [pc]        0
dtype: int64
_______________________Solar System_________________________
Discovery ID          0
Discovery Method      0
Discovery Year        0
Discovery Facility    0
dtype: int64
_______________________Discoveries__________________________
```

In [30]:
```
#store cleaned datasets in new CSV files
```

```python
#save stars data
cleaned_stars_data = stars_data.copy()
cleaned_stars_data.to_csv('cleaned_stars_data.csv', index=False)
print(cleaned_stars_data)

#save solar system data
cleaned_system_data = system_data.copy()
cleaned_system_data.to_csv('cleaned_system_data.csv', index=False)
print(cleaned_system_data)

#save discovery data
cleaned_discoveries_data = discoveries_data.copy()
cleaned_discoveries_data.to_csv('cleaned_discoveries_data.csv', index=False)
print(cleaned_discoveries_data)

#store cleaned exoplanet data
cleaned_planet_data = new_planet_data.copy()
cleaned_planet_data.to_csv('cleaned_planet_data.csv', index=False)
print(cleaned_planet_data)
```

```
     Star ID Spectral Type  Stellar Radius [Solar Radius]  \
0          1          G8 V                         0.9430
1          2          M1 V                         0.8620
3          4          G9 V                         1.0000
4          5        M2.5 V                         0.3026
5          6        M4.5 V                         0.2211
..       ...           ...                            ...
157      158            G8                         1.0500
158      159        M2.5 V                         0.4060
159      160          K0 V                         0.9900
160      161          G0 V                         1.4800
161      162          G0 V                         1.1700

     Stellar Mass [Solar mass]  Stellar Metallicity [dex]  \
0                       0.9050                      0.350
1                       0.6350                      0.010
3                       0.9500                     -0.100
4                       0.2952                     -0.070
5                       0.1945                     -0.170
..                         ...                        ...
157                     1.0600                      0.140
158                     0.4050                     -0.170
159                     0.9710                      0.430
160                     1.4050                      0.432
161                     1.0700                      0.090

     Stellar Luminosity [log(Solar)]  \
0                          -0.19695
1                          -0.97757
3                          -0.17000
4                          -1.96019
5                          -2.32148
..                              ...
157                         0.03226
158                        -1.62986
159                        -0.08823
160                         0.43132
161                         0.15836

     Stellar Surface Gravity [log10(cm/s**2)]  Stellar Age [Gyr]  \
0                                       4.430             5.500
1                                       4.370             0.019
3                                       4.400             0.600
4                                       4.840             9.900
5                                       5.037             5.000
..                                        ...               ...
157                                     4.420             8.000
158                                     4.870             4.100
159                                     4.433             7.800
160                                     4.246             1.180
161                                     4.430             2.980

     Stellar Density [g/cm**3]  Stellar Rotational Velocity [km/s]  \
0                     1.521203                                1.23
1                     1.750000                                9.20
3                     1.350000                                8.00
```

```
4                     1.620000                                2.00
5                    25.300000                                2.00
..                         ...                                 ...
157                   1.299000                                1.50
158                   8.540000                                2.00
159                   1.410000                                1.07
160                   0.451084                                6.00
161                   0.944584                                3.16

     Stellar Rotational Period [days]  Systemic Radial Velocity [km/s]  \
0                               38.80                        27.40949
1                                4.86                        -6.31031
3                                5.40                        29.55000
4                               30.00                        14.65000
5                              122.30                        35.07880
..                                ...                             ...
157                             20.16                        -1.23300
158                             44.40                         9.39000
159                             28.60                        46.93290
160                              3.62                       -12.98300
161                             18.30                        10.70916

     System ID
0            1
1            2
3            4
4            5
5            6
..         ...
157        158
158        159
159        160
160        161
161        162

[131 rows x 13 columns]
     System ID Host Name  Number of Stars  Number of Planets    RA [deg]  \
0            1    55 Cnc                2                  7  133.146837
1            2    AU Mic                1                  4  311.291137
3            4  CoRoT-18                1                  1   98.172417
4            5    G 9-40                1                  1  134.718851
5            6   GJ 1132                1                  2  153.709070
..         ...       ...              ...                ...         ...
157        158   WASP-96                1                  1    1.046570
158        159  Wolf 327                1                  1  148.378320
159        160    XO-2 N                2                  3  117.026769
160        161      XO-7                1                  1  277.478075
161        162  HD 39091                1                  3   84.299280

     Dec [deg]  Distance [pc]
0    28.329815        12.5855
1   -31.342450         9.7221
3    -0.031605       764.8890
4    21.074796        27.9276
5   -47.154936        12.6130
..         ...            ...
```

```
157 -47.360635       352.4640
158  35.570548        28.5321
159  50.225147       154.2730
160  85.233321       234.1490
161 -80.464604        18.2702

[130 rows x 7 columns]
     Discovery ID Discovery Method  Discovery Year  \
0               1  Radial Velocity          2004.0
1               2          Transit          2020.0
2               3          Transit          2021.0
3               4          Transit          2011.0
4               5          Transit          2019.0
5               6          Transit          2015.0
6               7          Transit          2009.0
7               8          Transit          2022.0
8               9  Radial Velocity          2004.0
9              10  Radial Velocity          2021.0
10             11          Transit          2017.0
13             14          Transit          2024.0
14             15          Transit          2008.0
15             16          Transit          2010.0
19             20          Transit          2007.0
20             21          Transit          2016.0
21             22          Transit          2016.0
24             25  Radial Velocity          2005.0
25             26  Radial Velocity          2019.0
28             29          Transit          2019.0
30             31  Radial Velocity          2007.0
34             35  Radial Velocity          1999.0
38             39          Transit          2016.0
42             43          Transit          2024.0
56             57          Transit          2018.0
58             59          Transit          2015.0
66             67          Transit          2017.0
67             68          Transit          2014.0
69             70          Transit          2011.0
70             71          Transit          2012.0
74             75          Transit          2016.0
75             76          Transit          2013.0
85             86          Transit          2017.0
89             90          Transit          2019.0
101           102          Transit          2023.0
105           106          Transit          2025.0
127           128          Transit          2008.0
128           129          Transit          2017.0
129           130          Transit          2016.0
130           131          Transit          2016.0
132           133          Transit          2016.0
135           136          Transit          2020.0
136           137          Transit          2019.0
137           138          Transit          2018.0
139           140          Transit          2019.0
141           142          Transit          2020.0
142           143          Transit          2009.0
143           144          Transit          2010.0
```

144	145	Transit	2007.0
145	146	Transit	2011.0
147	148	Transit	2012.0
151	152	Transit	2014.0
154	155	Transit	2015.0
159	160	Transit	2007.0
160	161	Transit	2019.0
161	162	Transit	2018.0

	Discovery Facility
0	McDonald Observatory
1	Transiting Exoplanet Survey Satellite (TESS)
2	Transiting Exoplanet Survey Satellite (TESS)
3	CoRoT
4	K2
5	Cerro Tololo Inter-American Observatory
6	MEarth Project
7	Transiting Exoplanet Survey Satellite (TESS)
8	W. M. Keck Observatory
9	Calar Alto Observatory
10	K2
13	Transiting Exoplanet Survey Satellite (TESS)
14	HATNet
15	HATNet
19	HATNet
20	HATSouth
21	Multiple Observatories
24	Haute-Provence Observatory
25	La Silla Observatory
28	Transiting Exoplanet Survey Satellite (TESS)
30	Subaru Telescope
34	W. M. Keck Observatory
38	K2
42	K2
56	K2
58	K2
66	KELT-North
67	Kepler
69	Kepler
70	Kepler
74	Kepler
75	Kepler
85	Qatar
89	Qatar
101	Transiting Exoplanet Survey Satellite (TESS)
105	Transiting Exoplanet Survey Satellite (TESS)
127	SuperWASP
128	SuperWASP-South
129	SuperWASP-North
130	SuperWASP
132	SuperWASP-South
135	SuperWASP-North
136	WASP-South
137	SuperWASP-South
139	SuperWASP
141	SuperWASP

```
142                                   SuperWASP
143                                   SuperWASP
144                                   SuperWASP
145                                   SuperWASP
147                                   SuperWASP
151                                   SuperWASP
154                                   SuperWASP
159                                          XO
160                                          XO
161  Transiting Exoplanet Survey Satellite (TESS)
     Planet ID Planet Name Planet Letter  Number of Moons  \
0            1    55 Cnc e             e                0
1            2    AU Mic b             b                0
2            3    AU Mic c             c                0
3            4  CoRoT-18 b             b                0
4            5    G 9-40 b             b                0
..         ...         ...           ...              ...
157        158   WASP-96 b             b                0
158        159  Wolf 327 b             b                0
159        160    XO-2 N b             b                0
160        161      XO-7 b             b                0
161        162    pi Men c             c                0

     Orbital Period [days]  Orbit Semi-Major Axis [au]  \
0                 0.736547                     0.01544
1                 8.463446                     0.07000
2                18.859023                     0.11900
3                 1.900069                     0.02950
4                 5.745998                     0.04180
..                     ...                         ...
157               3.425260                     0.04530
158               0.573474                     0.01000
159               2.615862                     0.03680
160               2.864142                     0.04421
161               6.267840                     0.06839

     Angular Separation [mas]  Planet Radius [Earth Radius]  \
0                      1.2300                      1.875000
1                      7.2000                      4.790000
2                     12.2000                      2.790000
3                      0.0386                     14.680000
4                      1.5000                      1.900000
..                        ...                          ...
157                    0.1290                     13.450000
158                    0.3500                      1.240000
159                    0.2390                     11.130537
160                    0.1890                     15.389957
161                    3.7400                      2.018900

     Planet Mass or Mass*sin(i) [Earth Mass]  Planet Density [g/cm**3]  ...  \
0                                    7.99000                     6.660  ...
1                                    8.99000                     0.490  ...
2                                   14.46000                     3.900  ...
3                                 1102.82000                     2.200  ...
4                                    4.00000                     3.200  ...
..                                       ...                       ...  ...
```

```
157                              152.55000                    0.370  ...
158                                2.53000                    7.240  ...
159                              179.89178                    0.715  ...
160                              225.34147                    0.340  ...
161                                3.63000                    2.800  ...

     Insolation Flux [Earth Flux]  Equilibrium Temperature [K]  \
0                       2657.8322                       1958.0
1                         21.2000                        554.8
2                          7.3000                        424.7
3                        823.7009                       1550.0
4                          6.2700                        440.6
..                            ...                          ...
157                      517.0000                       1285.0
158                      233.9000                        996.0
159                      428.0360                       1328.0
160                     1540.0000                       1743.0
161                      309.0000                       1169.8

     Inclination [deg]  Transit Midpoint [days]  Transit Depth [%]  \
0               83.590              2457063.210           0.038467
1               88.390              2458330.351           0.249000
2               89.460              2458342.223           0.085000
3               86.500              2455321.724           2.571701
4               89.030              2459503.327           0.365000
..                 ...                      ...                ...
157             85.600              2456258.062           1.380000
158             84.890              2459252.981           0.081839
159             88.010              2454508.738           1.336300
160             83.450              2457917.475           0.909000
161             87.045              2458425.789           0.026759

     Transit Duration [hours]  Radial Velocity Amplitude [m/s]  Star ID  \
0                    1.543900                            6.020        1
1                    3.240000                            3.800        2
2                    4.290000                            4.696        3
3                    2.387000                          590.000        4
4                    1.693000                            3.220        5
..                        ...                              ...      ...
157                  2.426400                           62.000      158
158                  0.900451                            3.540      159
159                  2.683900                           90.170      160
160                  2.772000                           80.500      161
161                  2.952024                            1.185      162

     System ID  Discovery ID
0            1             1
1            2             2
2            3             3
3            4             4
4            5             5
..         ...           ...
157        158           158
158        159           159
159        160           160
160        161           161
```

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161        162           162

[162 rows x 21 columns]
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